

Foundation & Phoenix

A technical overview of Nutanix's node imaging, cluster creation, and field provisioning tools.

What is Foundation?

Foundation is a Nutanix tool used to **image nodes** with a hypervisor (AHV, ESXi, Hyper-V) and Nutanix CVM, then **create a Nutanix cluster**.

It handles new nodes whether they arrive factory imaged, bare metal, or with Discovery OS installed.

Image Nodes

Create Clusters

Expand Clusters

Out-of-the-Box Node Configurations

When a customer receives new NX or OEM nodes, they arrive in one of three states:

1

Factory Imaged

Hypervisor on SATADOM/M.2 and AOS on first 1–2 SSDs. NX nodes always ship with AHV.

2

Bare Metal

No software installed. MITF (Meet in the Field) hardware always arrives in this state.

3

Discovery OS

Outside the USA, a lightweight OS is pre-installed to allow automatic node discovery via Foundation or Prism.

Three Foundation Flavors

CVM Foundation

Service running on CVMs. Requires factory-imaged nodes with hypervisor and CVM. Cluster creation is mandatory. Foundation service is normally **down** on configured clusters.

Standalone Foundation VM

Deployed on a Type II hypervisor. Discovers nodes via static IPMI IP or IPv6 link-local. Cluster creation is **optional** — supports staging nodes before on-site deployment.

Portable Foundation

Available as **.msi** (Windows) or **.dmg** (macOS). Bundled with a Type II hypervisor. Slightly limited vs. Standalone VM but covers most use cases.

Install Process: Factory Imaged Nodes

At minimum, one NIC per node must be connected. Best practice: connect both dedicated IPMI and shared ports for proper physical segmentation.

Foundation VM

Run on an external system on the same broadcast domain. Images all CVMs **in parallel** — faster method.

CVM-Based Foundation

Configure an IP on a running CVM, connect via `http://<IP>:8000/gui`.
Slower — imaging occurs in two phases. No separate VM required.

Bare Metal & Discovery OS Nodes

Bare Metal

Both the dedicated IPMI port and one hypervisor port must be connected. IPMI is used to image; hypervisor port configures and creates the cluster.

Discovery OS

Process is nearly identical to bare metal. The lightweight Linux OS enables automatic network discovery and supports **Software at Scale (S@S)** — allowing SREs to repair or rebuild nodes without a local Foundation copy.

What is Phoenix?

Phoenix is a tool used to **install the Nutanix CVM and provision a hypervisor** on a new or replacement node in the field. Each Phoenix ISO targets a specific hypervisor and AOS release.

Generated via Foundation

Follows the same versioning as Foundation (CVM or Foundation VM).

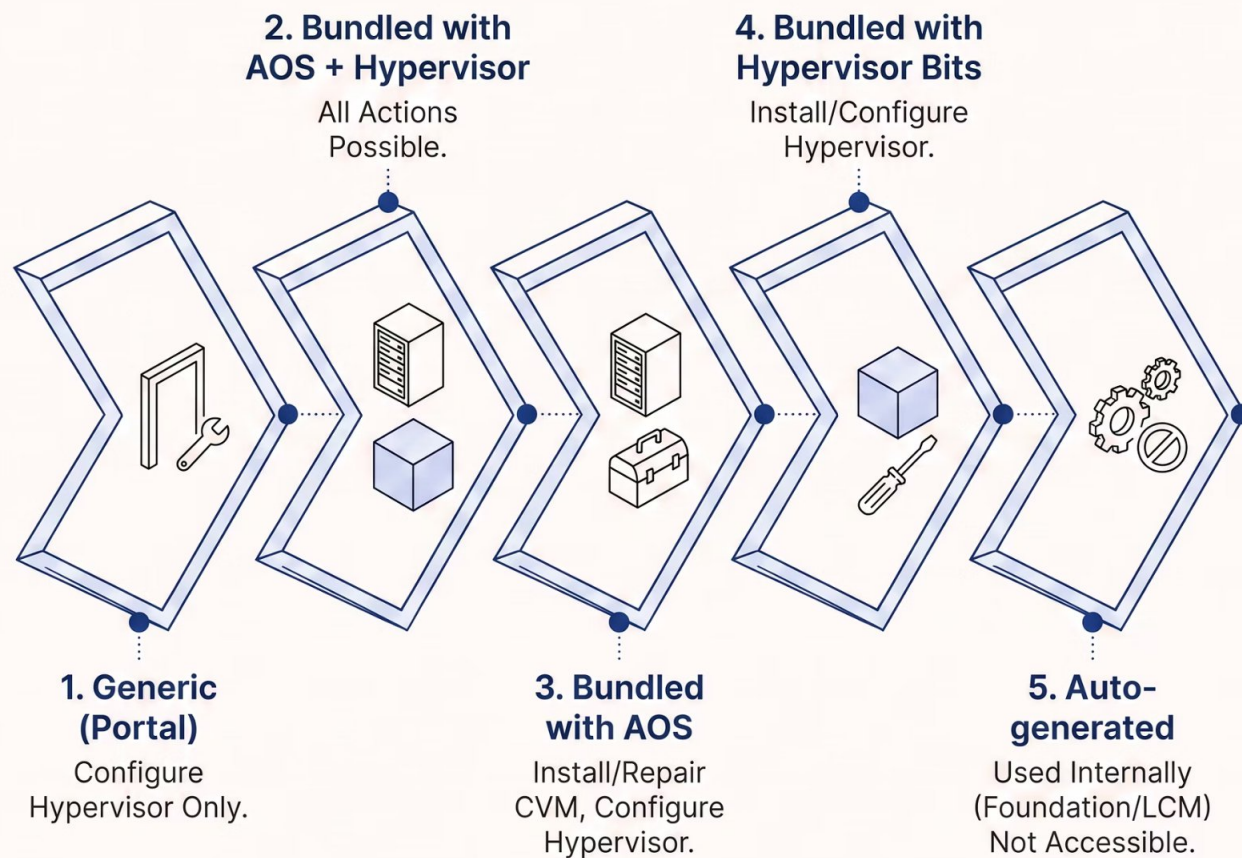
Version History

Versions prior to 4.3 were Gentoo-based; 4.3+ are CentOS-based.

Wide Usage

Used by support for CVM/hypervisor workflows, LCM firmware upgrades, SATADOM replacements, and more.

Phoenix ISO Types & Capabilities



Key Workflows Available

- Configure installed hypervisor
- Install CVM (wipes data) + install/configure hypervisor (*requires custom ISO*)
- Repair CVM (preserves data)
- Repair CVM + install/configure hypervisor (*requires custom ISO*)
- Install CVM only (*requires custom ISO with AOS bits*)



The portal ISO does **not** include AOS or hypervisor bits — custom-generated ISOs are required for full install workflows.

The "Light Phoenix" — Discovery OS

All Phoenix flavors include **NDPRescueShell mode** as the last boot menu option, enabling Phoenix to be discovered on the network.

Originally introduced for legal reasons — NX nodes shipping outside the US could not include AHV/AOS pre-installed. Bare metal foundation was required, a labor-intensive process.

US customs later permitted a lightweight discovery tool pre-installed on the hypervisor boot drive. Nodes boot, enter a **broadcasting/discovery state**, and can be imaged remotely by Foundation on a cluster, applet, or VM.

Phoenix Workflows & Troubleshooting

A comprehensive guide to Phoenix ISO workflows — covering configuration, installation, repair operations, ISO flavors, and common FATAL error resolution.

NUTANIX PHOENIX

The Four Primary Workflows

1

Configure Hypervisor

Uses vanilla Phoenix ISO. Pulls CVM info from SSDs. Host networking/name must be reconfigured.

2

Install Hypervisor

Requires hypervisor bits in Phoenix. Installs on boot drive and pulls CVM info from SSDs.

3

Install CVM

Requires AOS bits. Fresh CVM install — **wipes all disks** on the node. Does not preserve data.

4

Repair CVM

Requires AOS bits. Rebuilds root (/) and /home partitions. Data partition and other disks remain intact.

Phoenix ISO Flavors

Vanilla / Generic

Available on Nutanix Support Portal. Only enables **Configure Hypervisor**. Used for SATADOM/M2 failures and troubleshooting.

Phoenix + Hypervisor

Enables Install & Configure Hypervisor. Ideal for reimaging AHV nodes after SATADOM replacement — no second ISO needed.

Phoenix + AOS

Enables Install CVM, Repair CVM, Configure Hypervisor. Ideal for installing or rescuing a CVM on a node.

Phoenix + Hypervisor + AOS

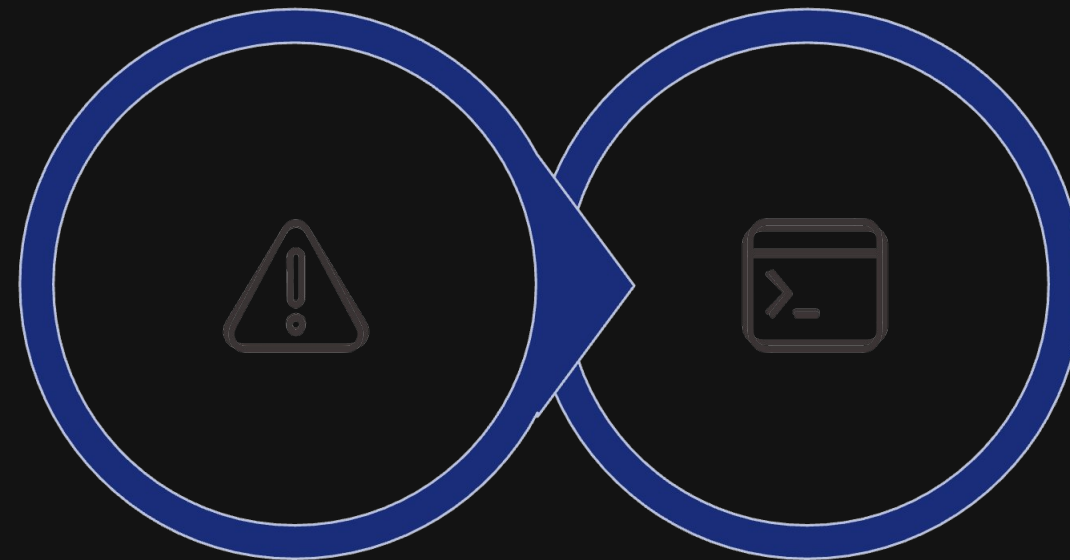
All four workflows available. Ideal for **fresh installation** of AOS and hypervisor on a node not yet in the cluster.

ISO Flavor → Workflow Matrix

ISO Flavor	Configure HV	Install HV	Install CVM	Repair CVM
Vanilla / Generic		×	×	×
Phoenix + Hypervisor			×	×
Phoenix + AOS		×		
Phoenix + HV + AOS				

When Phoenix Hits a FATAL

If a node booted into Phoenix encounters a FATAL, the GUI workflow selector will not appear — you'll drop into the Phoenix shell instead. Below are the most common FATALs and their fixes.



FATAL on
boot

Enter
Phoenix shell

Each FATAL has a specific cause and targeted fix. The following cards break down each error.

FATAL: AOS & Hypervisor Not Found

Error Message

"Couldn't find a hypervisor and an AOS installed. Please package phoenix with a hypervisor and AOS."

Cause

AOS and hypervisor not detected on SSDs or boot drive. Node booted into vanilla Phoenix — no workflows can start.

Fix

Package Phoenix with Hypervisor/AOS, or install hypervisor manually then use a Phoenix+AOS ISO to bring up the CVM.

FATAL: CVM Not Found on System

"Couldn't find a CVM installed. Please package an AOS with Phoenix."

Cause: AOS not on SSDs. Hypervisor installed and customized (.firstboot or .firstboot_success present). ISO has no AOS.

Fix: Package Phoenix with AOS and reimage, or install hypervisor then use Phoenix+AOS to install CVM.

"No CVM found. Firstboot for hypervisor failed. Please package Phoenix with hypervisor and AOS."

Cause: AOS not on SSDs. Hypervisor installed but firstboot failed (.firstboot_fail marker present). Vanilla ISO — no workflows available.

Fix: Package Phoenix with Hypervisor+AOS and reimage the node.

FATAL: Hypervisor Missing or Failed

"Couldn't find a hypervisor packaged with Phoenix. No hypervisor detected."

Cause: Hypervisor not installed; ISO has no hypervisor packaged.

Fix: Package Phoenix with Hypervisor, or install manually then use Phoenix to configure.

"The existing hypervisor installation is in a failed state. Please reinstall."

Cause: Hypervisor installed but firstboot failed (.firstboot_fail marker). ISO has no hypervisor.

Fix: Package Phoenix with Hypervisor and reimage, or install manually then configure via Phoenix.

FATAL: Hypervisor Customization Issues

"Hypervisor already customized. Maybe run firstboot?"

Cause: Hypervisor installed and customized (.firstboot marker present) but firstboot hasn't completed. ISO has no hypervisor packaged.

Fix: Try running firstboot again. If unsuccessful, package Phoenix with Hypervisor or install manually, then configure via Phoenix.

"Hypervisor and AOS already installed. Nothing to do."

Cause: Both AOS and Hypervisor are installed and configured. Node booted into generic vanilla Phoenix.

Fix: Package Phoenix with AOS and/or Hypervisor and select the appropriate workflow.

Quick Reference: FATAL → Fix

FATAL Condition	Root Cause	Resolution
AOS & Hypervisor not found	Vanilla ISO, nothing on SSDs/boot drive	Package Phoenix with HV+AOS
CVM not found (customized HV)	No AOS on SSDs, HV customized	Package Phoenix with AOS, reimage
CVM not found + firstboot failed	No AOS, .firstboot_fail marker	Package Phoenix with HV+AOS
No hypervisor detected	HV not installed, no HV in ISO	Package Phoenix with HV
HV customized, firstboot pending	.firstboot marker, no HV in ISO	Run firstboot or reimage with HV
HV in failed state	.firstboot_fail, no HV in ISO	Package Phoenix with HV, reimage
HV & AOS already installed	Vanilla ISO, nothing to do	Package Phoenix with AOS/HV